



NTNU – Trondheim
Norwegian University of
Science and Technology

TET4185: Power Markets, Resources and Environment

Participating in Electricity Markets

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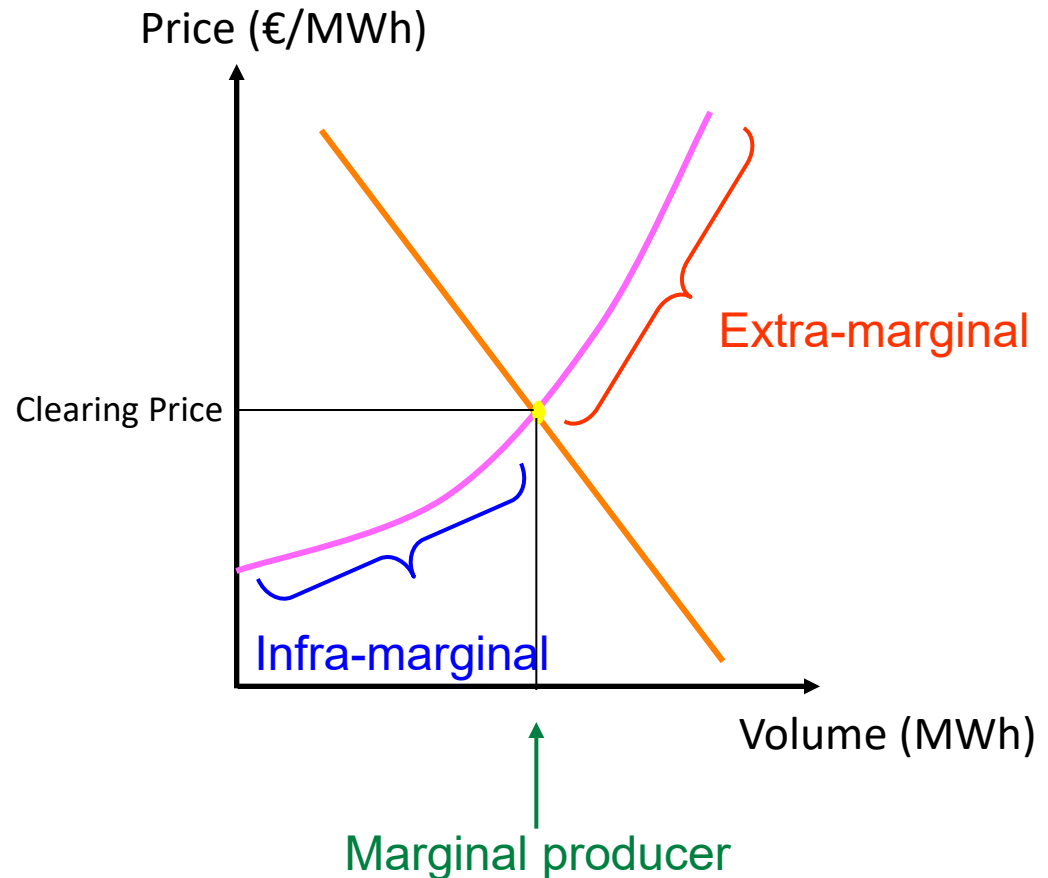
Perspectives for participating in electricity markets

- Generator
- Consumer
- Retailer

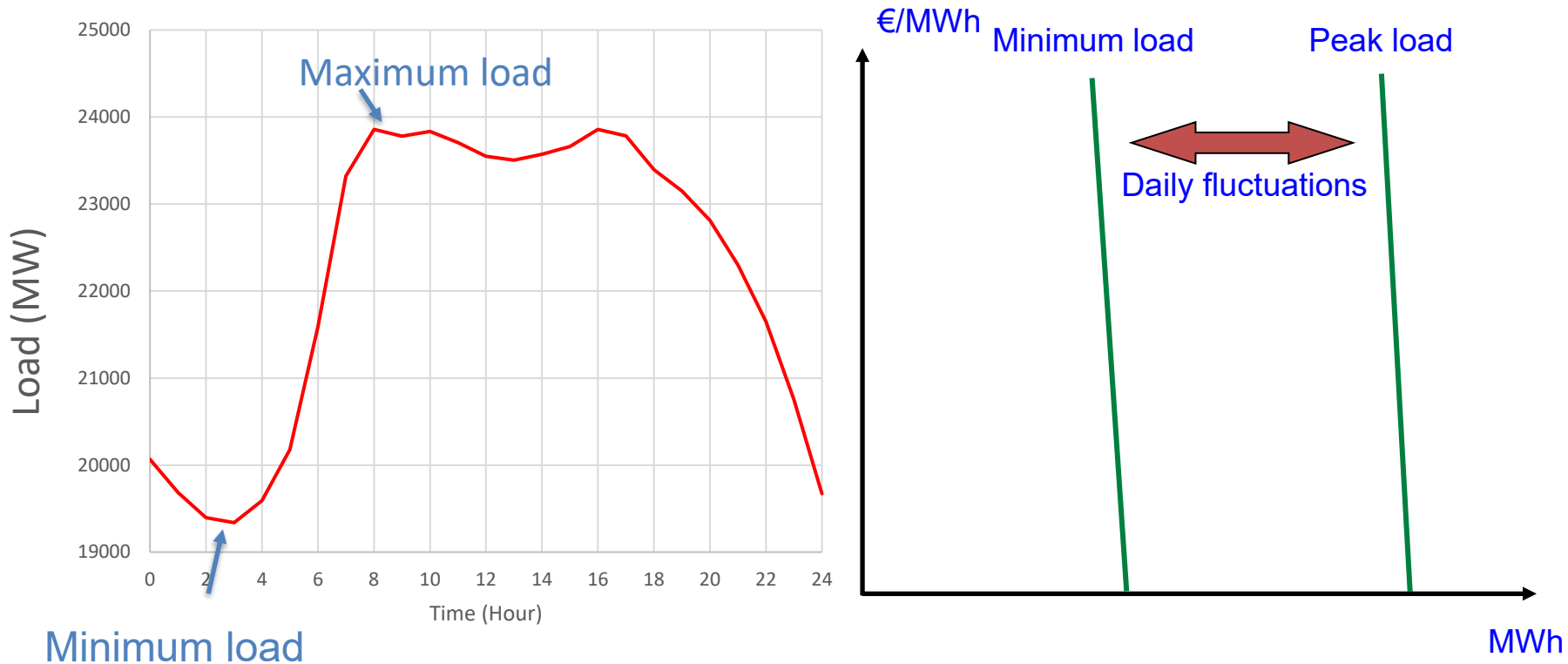
Participating in Electricity Markets: The Generator's Perspective

Marginal, infra-marginal, extra-marginal producers

- Everything is sold at the market clearing price. Price is set by the “last” unit sold
- **Marginal** producer:
 - Sells this last unit
 - Gets exactly its bid
- **Infra-marginal** producers:
 - Get paid more than their bid
 - Collect economic profit, “Producer surplus”
- **Extra-marginal** producers:
 - Sell nothing

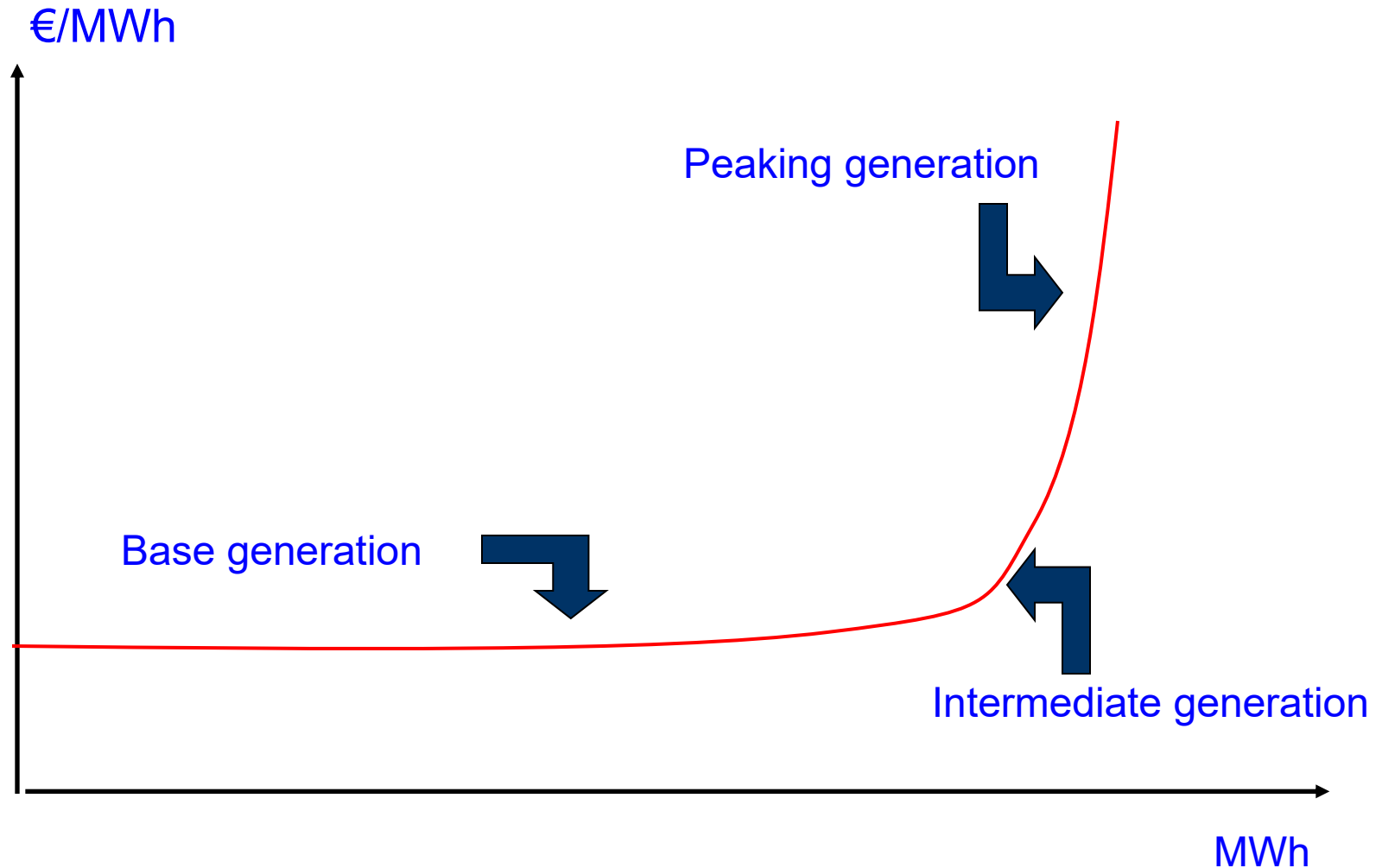


Demand curves for electricity

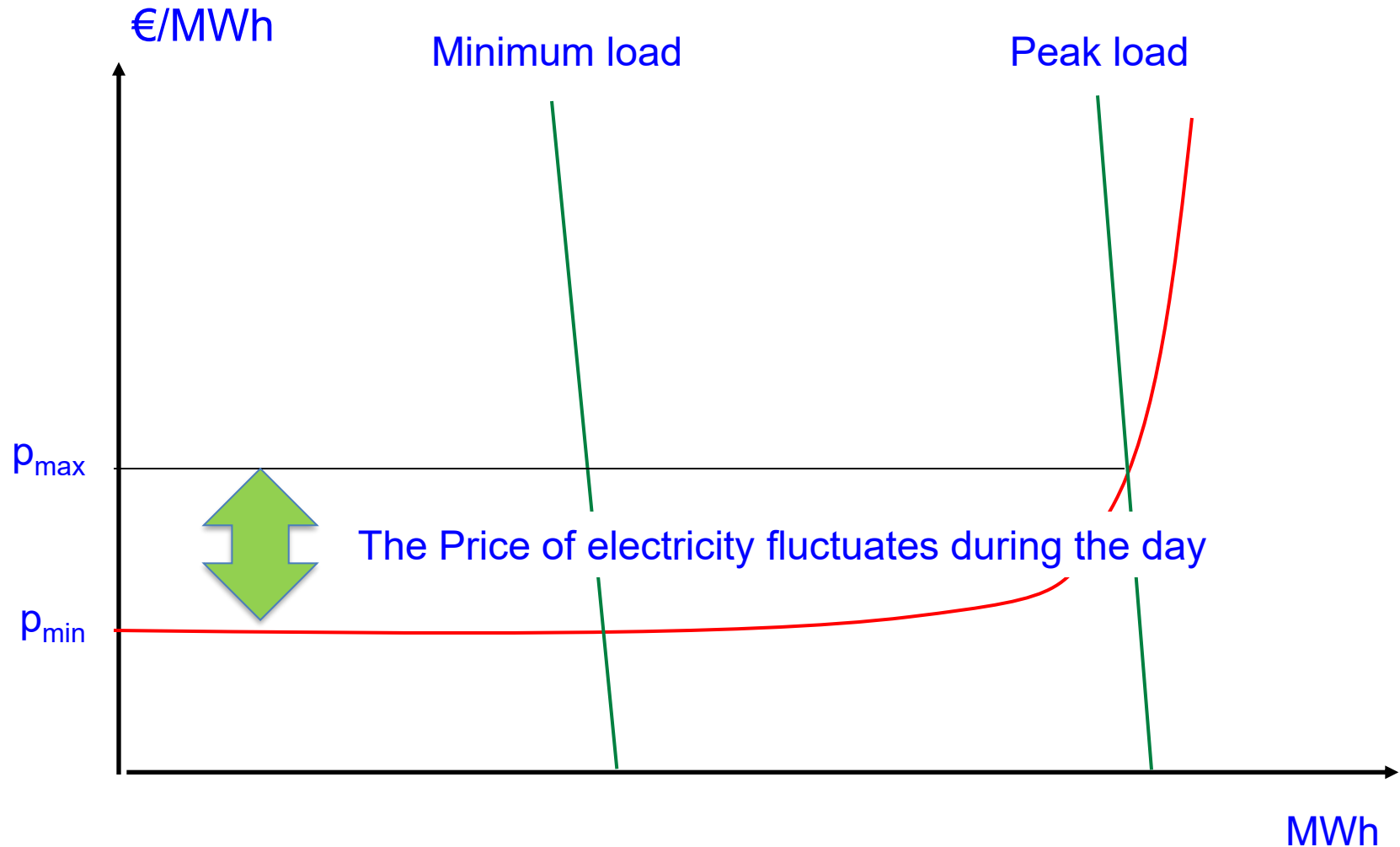


Electricity consumption in Norway on 07.01.2016, Source: NordPool

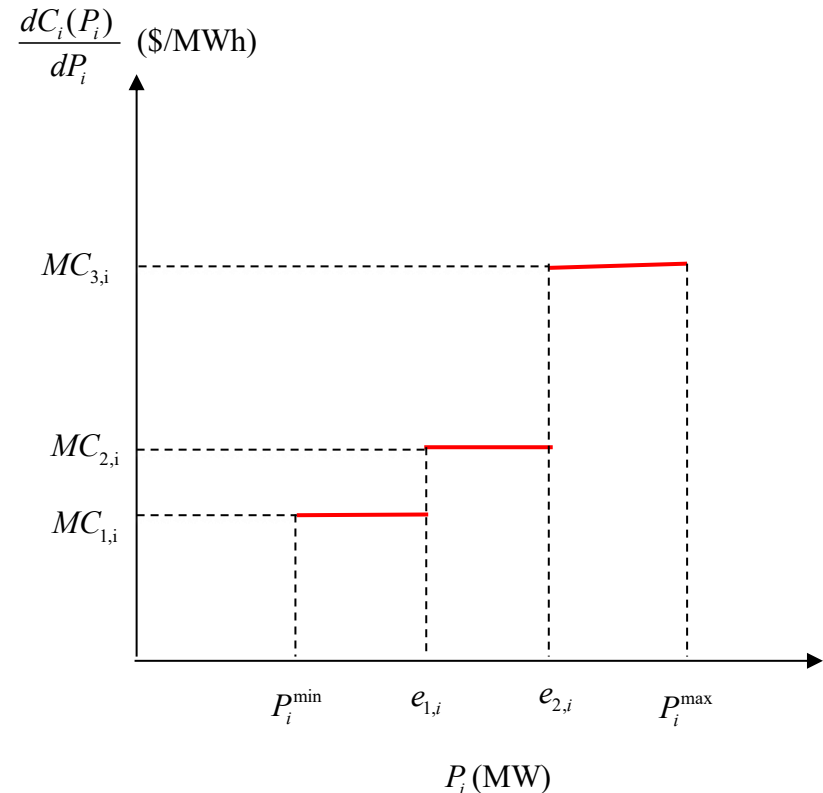
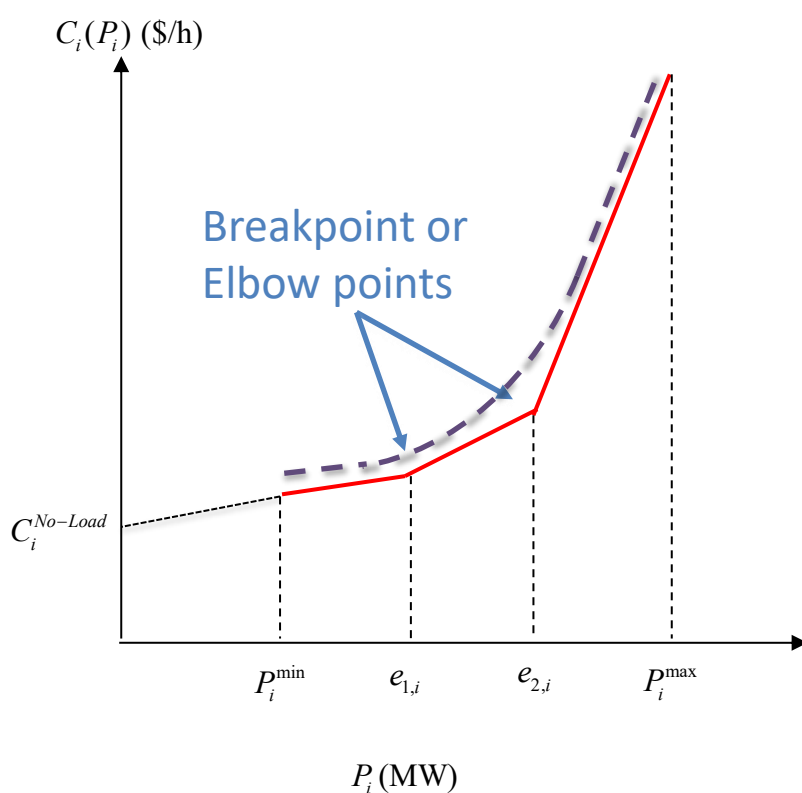
Supply curve for electricity



Supply and demand for electricity



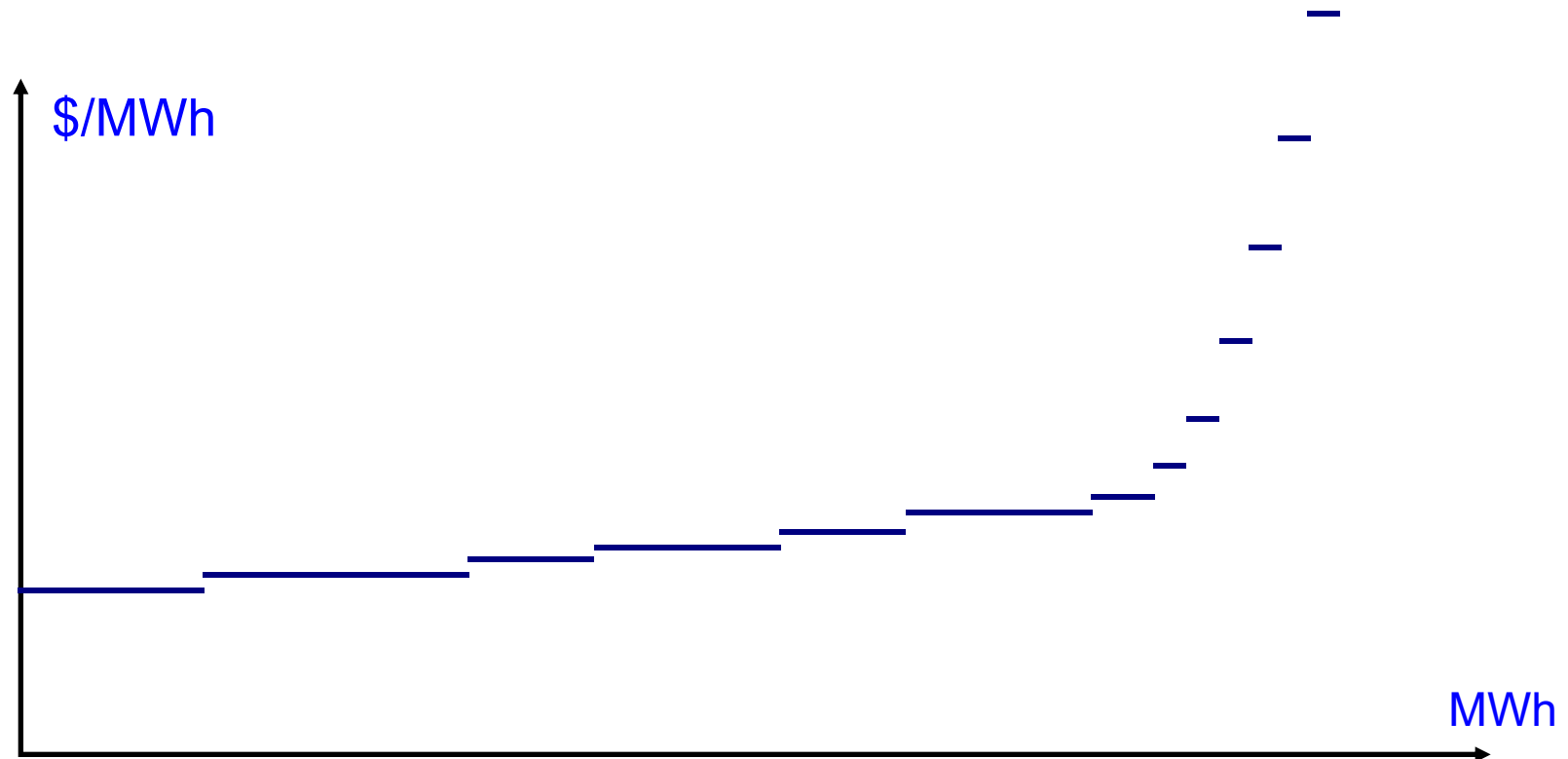
Piecewise linear cost curves



- **Piecewise Linear Representation:** The cost curve is segmented into linear parts, each representing a specific operational range of a power plant.
- Left graph shows the total cost curve ($C(P)$) as a function of power output (P), highlighting the non-load and variable cost components.
- Right graph shows the derivative of the cost function ($\frac{dC}{dP}$), or the marginal cost, as a function of power output, divided into segments (MC_1, MC_2, MC_3) representing different operational stages.

Supply curve for electricity

- In a centralized market, the supply curve is built by ranking the offers made by the generators
- A bid specifies the quantity that the generator is willing to sell at a given price



Bidding in a centralised market

- How should a generator bid to maximize its profit?

It depends on how much competition it has!

Market Structure



- Monopoly:
 - Monopolist sets the price at will
 - Must be regulated
- Perfect competition:
 - No participant is large enough to affect the price
 - All participants act as “price takers”
- Oligopoly:
 - Some participants are large enough to affect the price
 - Strategic bidders have market power
 - Others are price takers

Bidding into the market (perfect competition)

- Generating Company tries to maximise the profit it drives from the sale of electrical energy:

$$\text{Max } \pi_i = \text{Max} \left[\underbrace{p \cdot q_i}_{\text{Revenue}} - \underbrace{C_i(q_i)}_{\text{Production cost}} \right]$$

$$\frac{d\pi_i}{dq_i} = \frac{d(p \cdot q_i)}{dq_i} - \frac{dC_i(q_i)}{dq_i} = 0 \Rightarrow MR_i = MC_i$$

- In **perfect competition**, the price is not effected by quantity
→ the generators are price-taking

$$MR_i = \frac{d(p \cdot q_i)}{dq_i} = p \Rightarrow \frac{dC_i(q_i)}{dq_i} = p$$

- Adjust production q until the marginal cost of production is equal to the price p
- The marginal cost includes the cost of fuel, maintenance and all other items that vary with the power produced by one unit

Example-Perfect Competition

- Consider a coal-fired steam unit whose minimum stable generation is 100 MW (i.e. the minimum amount of power that it can produce continuously) and whose maximum output is 500 MW. On the basis of measurements taken at the plant, the input-output curve of this unit is estimated as:

$$H(P) = 110 + 8.2 P + 0.002 P^2 \text{ MJ/h}$$

- If we assume that the cost of coal is 1.3 \$/MJ, and the price at which electrical energy can be sold is 12 \$/MWh, what is the optimum bid in perfect competition market?
- At which price level of electrical energy, the unit can operate at its maximum output?

Example-Perfect Competition

- If the cost of fuel is “ F ”, then

$$C_1(P_1) = 110 \times F + 8.2 \times F \times P_1 + 0.002 \times F \times P_1^2 \text{ \$}/\text{h}$$

- we assume that the cost of coal is 1.3 \$/MJ, the cost curve is:

$$C_1(P_1) = 143 + 10.66P_1 + 0.0026P_1^2 \text{ \$}/\text{h}$$

- The market price is 12 \$/MWh and the optimum bid in perfect competition market

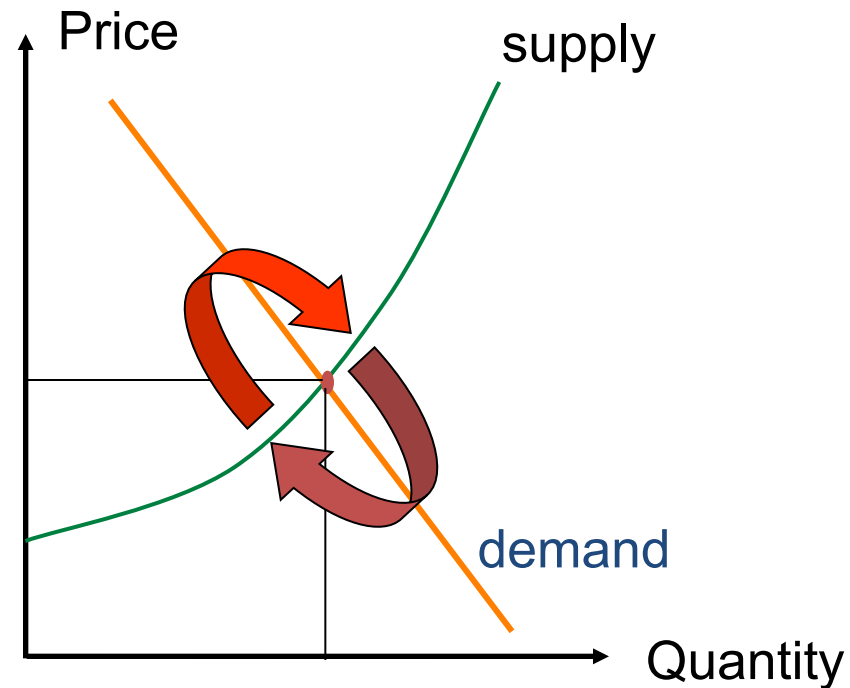
$$\frac{dC_1(P_1)}{dP_1} = 10.66 + 0.0052P_1 = 12 \text{ \$}/\text{MWh} \quad \text{or } P_1 = 257.7 \text{ MW}$$

- The unit can operate at its maximum output, whenever the price of electrical energy is greater than or equal to:

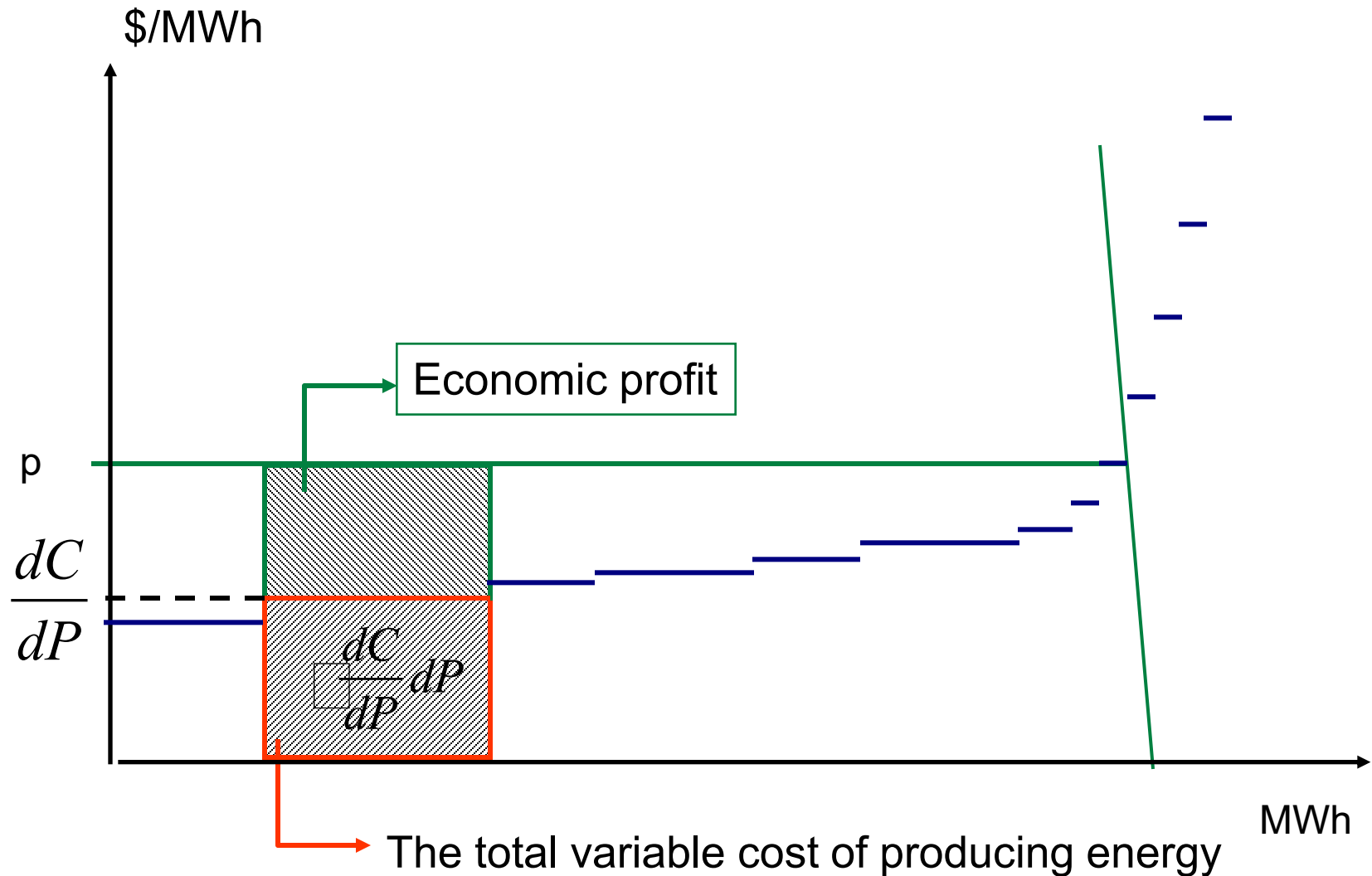
$$\left. \frac{dC_1(P_1)}{dP_1} \right|_{500 \text{ MW}} = 10.66 + 0.0052 \cdot 500 = 13.26 \text{ \$}/\text{MWh}$$

Bidding under perfect competition

- Since there are lots of small producers, a change in bid causes a change in the order of the bids
- If I bid at my marginal cost
 - I get paid the market clearing price if marginal or infra-marginal producer
- If I bid higher than my marginal cost
 - I could become extra-marginal and miss an opportunity to sell at a profit
- If I bid lower than my marginal cost
 - I could have to produce at a loss
- No incentive to bid anything else than marginal cost of production



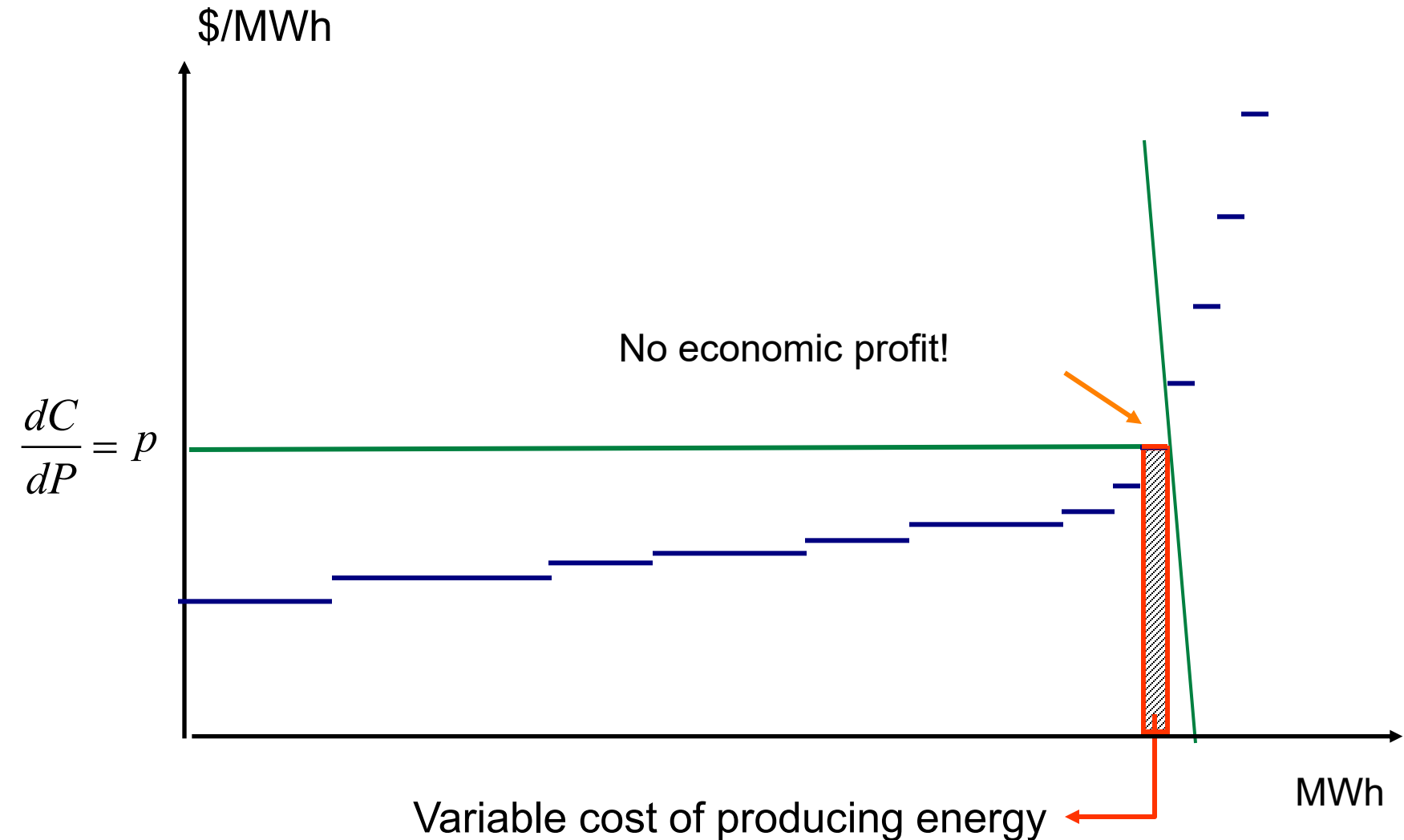
Profit of an infra-marginal producer



Profit of an infra-marginal producer

- Selling at marginal cost covers the **variable** cost of production
- The difference between the market price and the marginal cost must pay for the **fixed** costs:
 - No-load cost, start-up cost
 - Cost of building the plant
 - Interest payments for the bank, dividends for the shareholders
- A plant must therefore be infra-marginal often enough to cover its fixed costs
 - Market clearing price > marginal cost for enough hours of the year

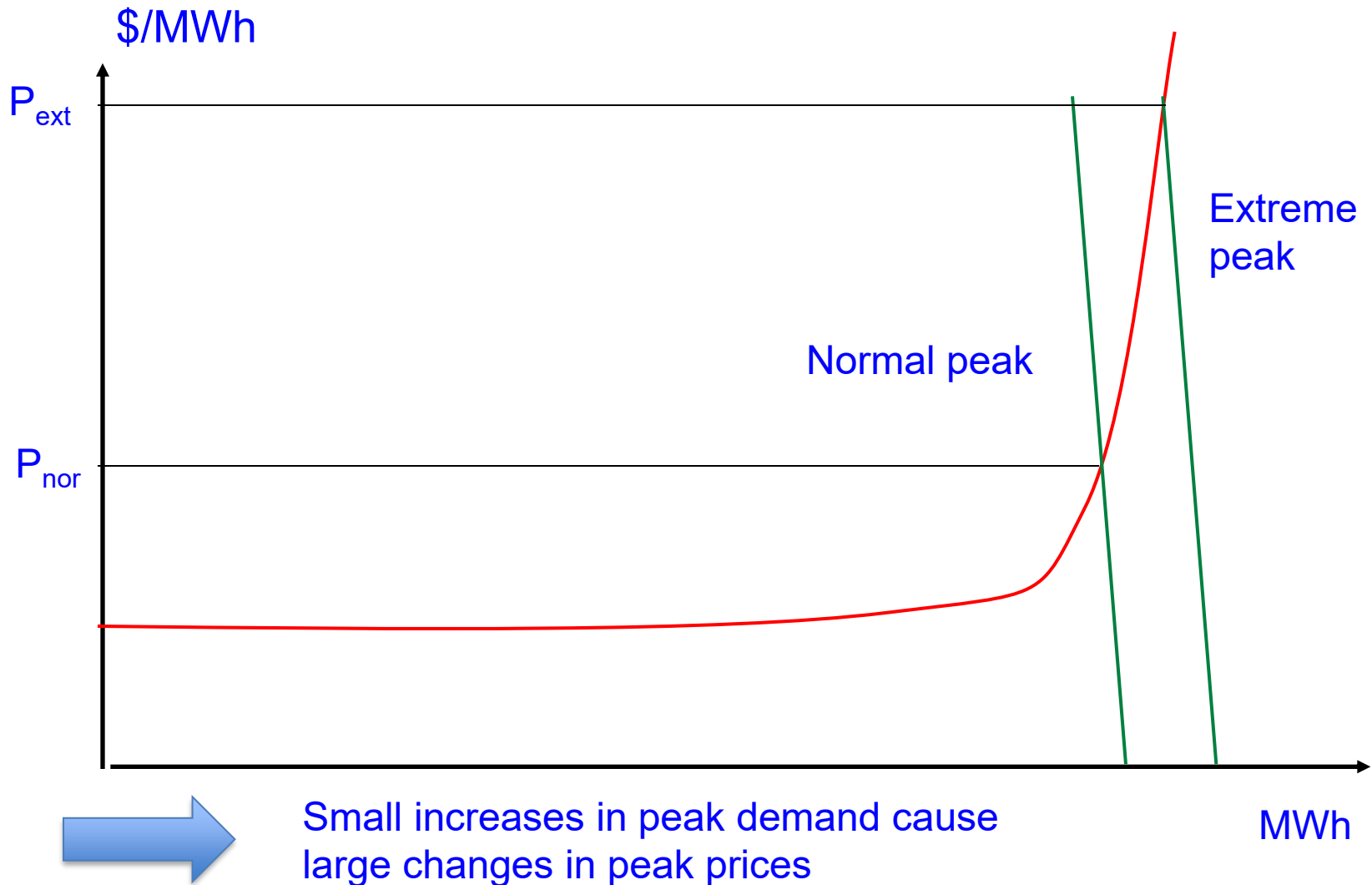
Profit of a marginal producer



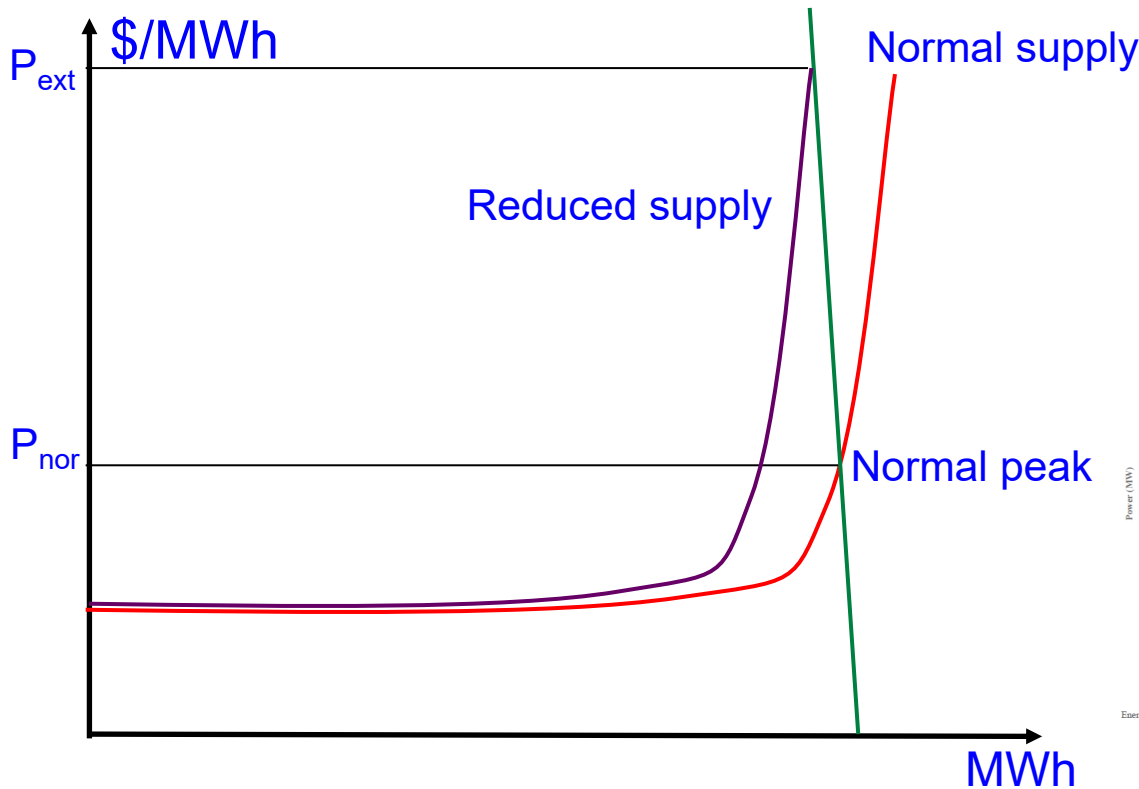
Profit of a marginal producer

- If a marginal generator bids at its marginal cost, it makes no economic profit
 - Covers only its variable cost of production
 - Does not cover its fixed cost
- Generators that are too often marginal or just below marginal will not recover their fixed costs if they bid at their marginal cost of production
 - They must include part of their fixed costs in their bid price
 - Their offer price is therefore higher than their marginal cost
 - They can do it because competition is not perfect when the load is high because most generators are already producing

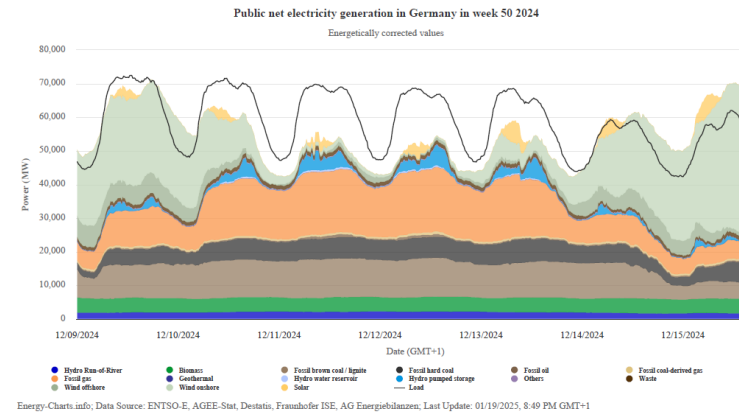
Price spikes because of increased demand



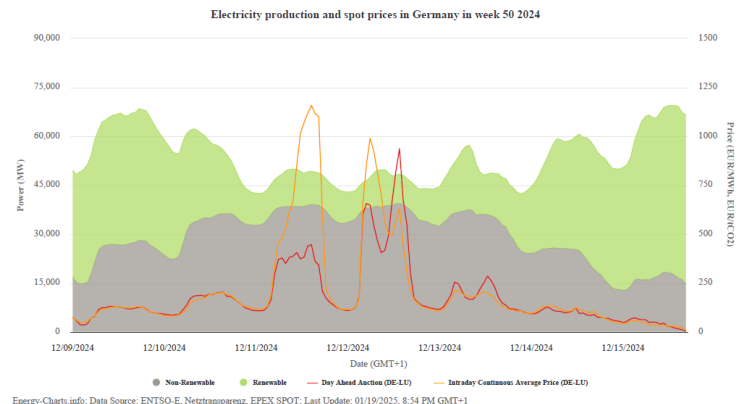
Price spikes because of supply scarcity



“Dunkelflaute” refers to a period during which there is little sunlight (“dunkel”=dark) as well as little wind (“Flaute”=lull, period of low wind speeds).



Graph from <https://energy-charts.info/>



Small reductions in supply cause large changes in peak prices

Optimal Scheduling

- Start-up cost
- Dynamic constraints
- Environmental constraints



Start-up cost

- The cost of getting the unit running and ready to produce from a shutdown state
- To maximise the profitability of a thermal units, start-up costs must be amortized over a period (bidding blocks)
- This may even involve operating the unit at a loss for a few hours than shutting it down

Example: Start-up cost

- Find an optimal schedule for the coal-fired plant in previous example over a period of seven hours.

$$C(P) = 143 + 10.66 P + 0.0026 P^2 \text{ \$/h}$$

$$P_{\min} = 100\text{MW} , MC_{\min} = 11.18\text{\$/MWh}$$

$$P_{\max} = 500\text{MW} , MC_{\max} = 13.26\text{\$/MWh}$$

- Assume that the price at which electrical energy can be sold is set on an hourly basis

Hour	1	2	3	4	5	6	7
Price (\\$/MWh)	13	13	14	10	12	14	12

- Suppose also that the generating unit is started up at hour 1 and that the cost of bringing it online is \$600.

Solution

$$C(P) = 143 + 10.66 P + 0.0026 P^2 \text{ \$/h}$$

$$P_{\min} = 100\text{MW} , MC_{\min} = 11.18\$/\text{MWh}$$

$$P_{\max} = 500\text{MW} , MC_{\max} = 13.26\$/\text{MWh}$$

Hour	1	2	3	4	5	6	7
Price (\$/MWh)	13	13	14	10	12	14	12
Generation (MW)	450.0	450.0	500.0	100.0	257.7	500.0	257.7
Revenue (\$)	5850	5850	7000	1000	3092	7000	3092
Running cost (\$)	5467	5467	6123	1235	3063	6123	3063
Start-up cost (\$)	600	0	0	0	0	0	0
Total cost (\$)	6067	5467	6123	1235	3063	6123	3063
Profit (\$)	-217	384	877	-235	30	877	30
Cumulative profit (\$)	-217	167	1044	809	839	1716	1745

Temporal Constraints

- Ramp rate constraints: Excessive stress damages the plant and shortens its life
- Minimum uptime and minimum downtime: These limits ensure that there is enough time for the temperature gradient in the turbine to subside

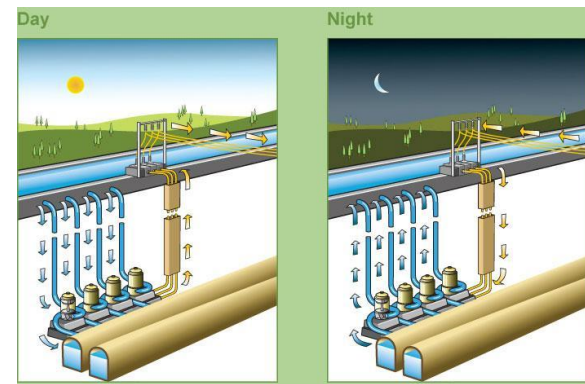
Environmental Constraints

- Constraints on amount of CO₂ emission from thermal power plans
- Constraints on use of hydro
 - ecological constraints
 - reservoir level

Participating in Electricity Markets: The consumer's perspective

Short-term Behaviour of Consumer

- Normal consumers pay the flat rate for each kWh
- Their demand reflects only their willingness to pay this rate
- The price elasticity of the demand for electricity is small
- Rather than response to sudden increase in the electricity prices, consumers decide to delay this demand until a time when prices are low



Options for the consumers

- Buy at the spot price
 - Lowest cost, the highest risk
 - Requires sophisticated control of the load
 - Must be managed carefully and requires sophisticated control of the load
- Buy from a retailer at a tariff linked to the spot price
 - Retailers acts as an intermediary between consumer and market
 - Risk can be limited by placing a cap on the price
- Interruptible contract
 - Reasonable option only if the cost of interruption is not too high
 - Savings can be substantial

Options for the consumers

- Buy from a retailer on a time-of-use tariff
 - Shifts some of the risk to the consumer
 - Need to control the load to save money
- Buy from a retailer at a fixed tariff
 - Lowest risk, highest cost
 - Two components to the price: average cost of energy and risk premium

Norgespris

- Government proposal: Norgespris (fixed 50 øre/kWh including VAT)
- Consumption caps:
 - 5,000 kWh/month for households
 - 1,000 kWh/month for cabins
- Duration: Oct 2025 – Dec 2026
 - Valid for consumption within the cap
 - Usage above cap billed at market price
 - Standard grid tariffs & surcharges still apply
- Aim: predictability & security for households.



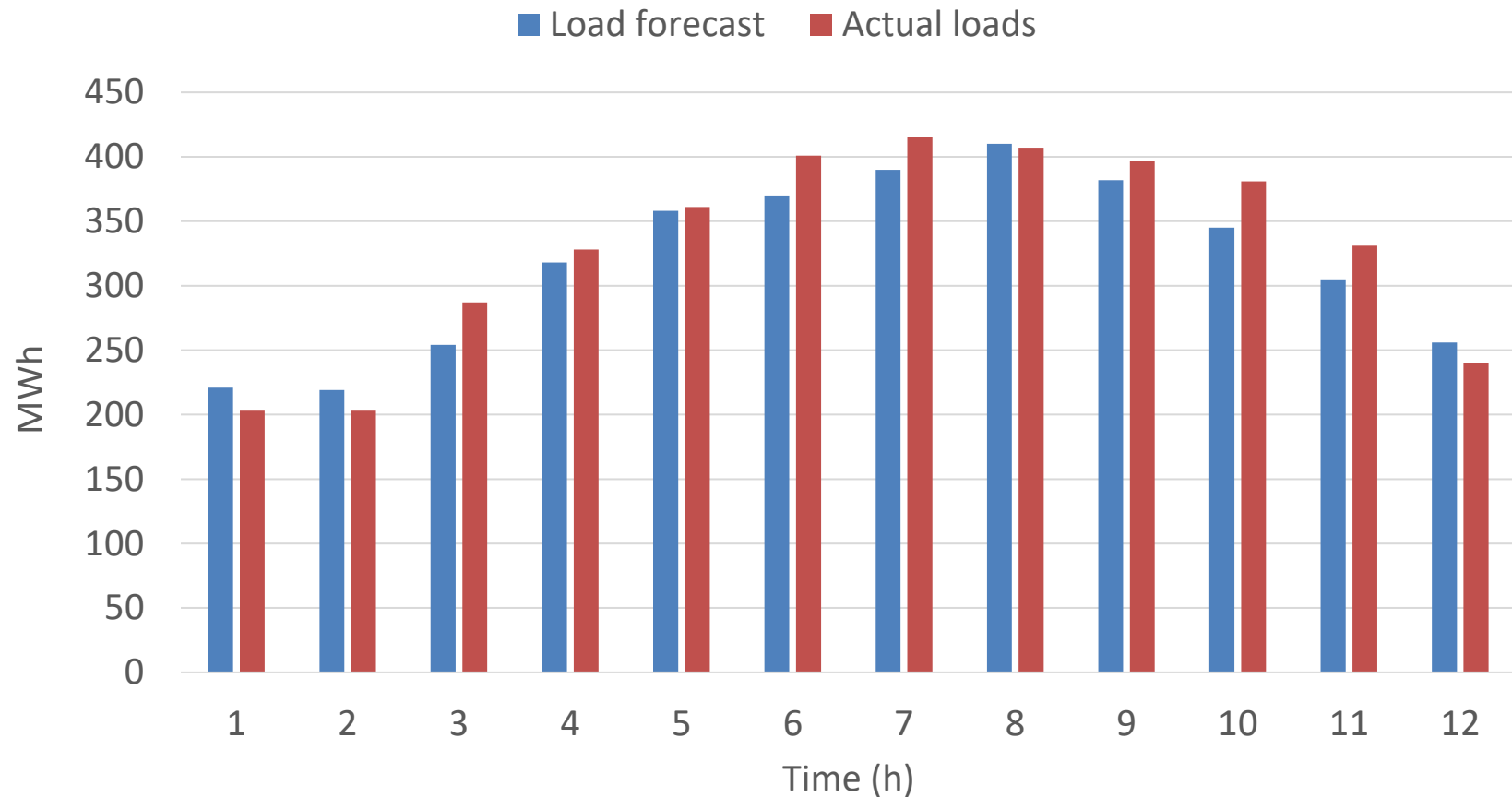
<https://www.nve.no/reguleringsmyndigheten/kunde/stroem/dette-er-norgespris/>

Participating in Electricity Markets: The retailer's perspective

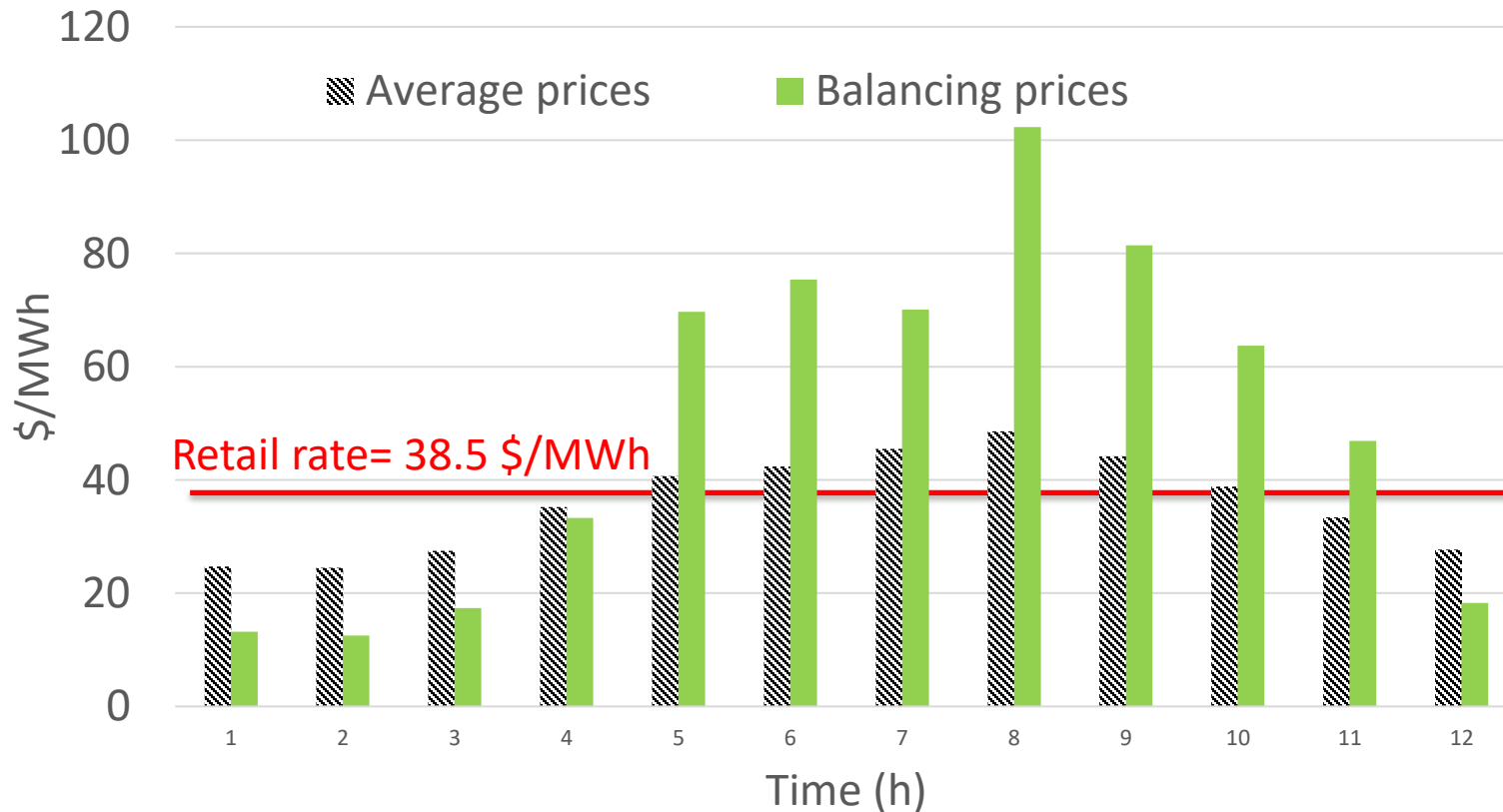
The retailer's perspective

- Sell energy to consumers, mostly at a flat rate
- Buy energy at a variable price on the whole sale market
 - Spot price
 - Contracts
- Must forecast the load of its customers and participate in wholesale market based on their forecast
- Want to reduce risks associated with market price
- The quantity-weighted average price at which they purchase energy should be lower than the rate they charge their consumers

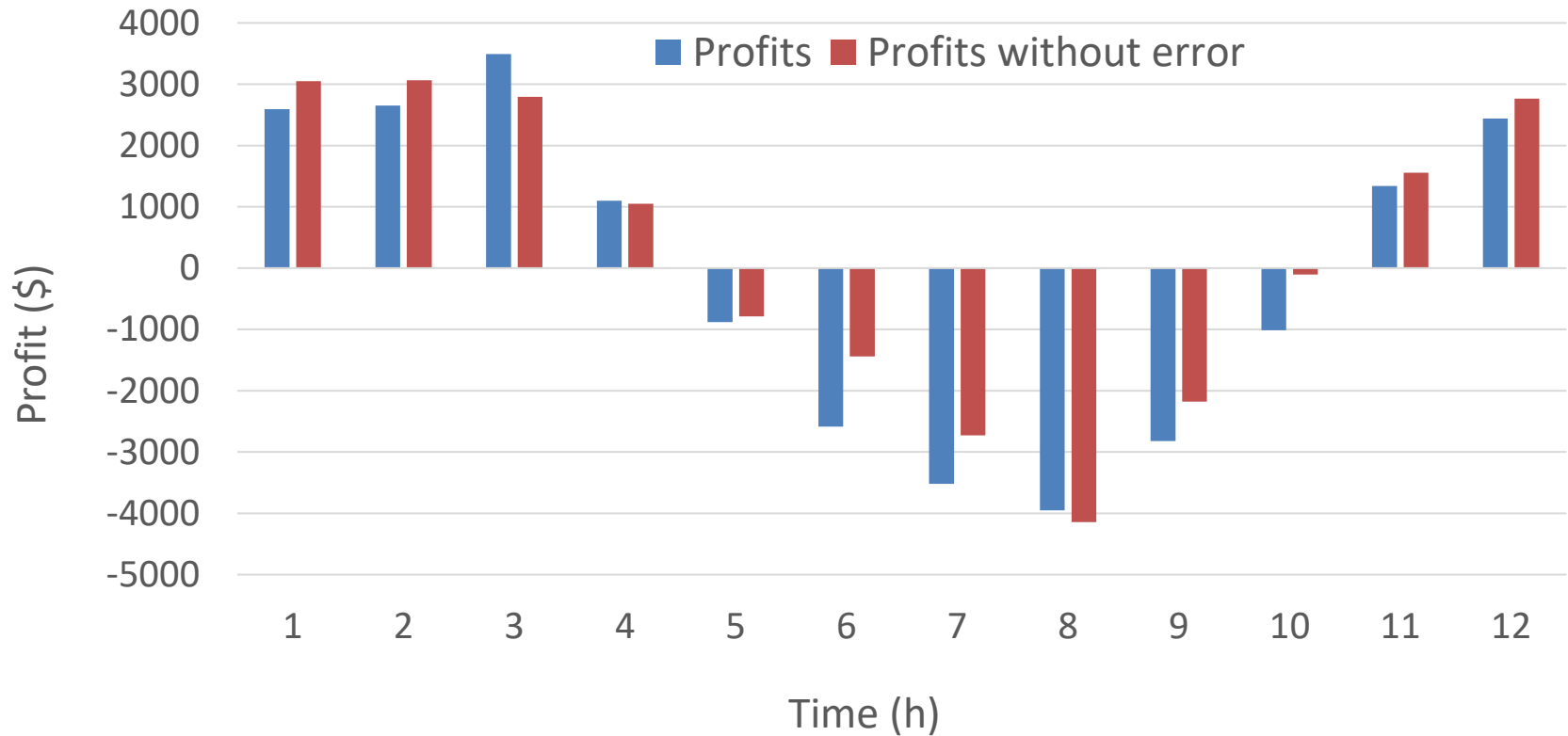
Example- Load Forecast and Actual loads



Example- Average Price and Balancing Prices



Example-Profits



Total Profits = \$ -1154
Profits without error = \$ 2896